Table of Content

Table of Content 1
Table of Figures 1
Background 2
Research Goals 2
Definitions 2
Research Outputs in Israel 3
Findings and Insights Based on In-Depth Interviews 8
Israel's Strengths in Artificial intelligence, Data Science and Smart Robotics 12
Next Phases of the Research 12
Main Insights from Other National Strategic Plans 12
Literature Review Findings Related to Scientific and Technological Workforce Requirements 16
Sector Mapping: Government, Academia, Defense and Industry in Israel 19

Table of Figures

Figure 1: Total publications and publications-per-capita by country, 2013-2017 4
Figure 2: Changes in the number of Israeli publications[2] compared with 'comparable countries' 5
Figure 3: Number of internationally coauthored publications, 2008-2017 5
Figure 4: Number of publications[3] and normalized average citations in Israel and in 'comparable countries' [3], 2003-2017 6
Figure 5: Publications in 5 leading conferences by country, 2013-2017 7
Figure 6: Patent applications in AI for Israeli applicants, Israeli inventors and non-Israeli applicants (at least one Israeli inventor) 8
Figure 7: University researchers in Israel by institute 21
Figure 8: Research interests of Israeli researchers in the fields of artificial intelligence, data science and smart robotics 22
Figure 9: Second- and third-degree theses, 2007-2016 22
Figure 10: Israeli artificial intelligence companies by year of establishment 24

Background

The Israeli government acknowledges the potential of the Artificial Intelligence, Data Science and Smart Robotics domains in fostering the Israeli economy, its security and societal welfare. The government is interested in promoting and augmenting the development of these fields, by supporting private and public investments.

The National Council for Research and Development (MOLMOP) at the Ministry of Science and Technology has issued a tender requesting a study and review of the activities in these technological domains in Israel. The Samuel Neaman Institute was commissioned to perform a comprehensive mapping of activities in the Israeli academy, industry and government sectors, and to explore the possibilities for promoting and developing these fields in Israel.

This interim report constitutes the first output in a series of three deliverables planned in the framework of this project. The report summarizes the first eight months of the project, in which extensive work was carried out to define the respective research domains and to review and map the activities in these domains in Israel. The report also includes an initial identification of the strengths and weaknesses of the Israeli ecosystem in the AI, Data Science and Smart Robotics domains and offers preliminary Insights towards the formulation of a national strategic plan in these fields.

The literature survey in the appendix of the report, reviews among other topics, global strategic programs in these areas. A part of the literature survey is devoted to human capital and examines various skill and education requirements of local labor markets in selected countries.

Research Goals

The research goals are as follows:

- To map the activities in the fields of Artificial Intelligence, Data Science and Smart Robotics in the academia and in the industry in Israel; To review selected activities in the defense and the government sectors;
- To examine Israel's strengths and weaknesses and to identify opportunities and threats in these fields;
- To review strategic plans in leading countries in these fields;
- To present preliminary insights towards the formulation of a national strategic plan in these fields.

Definitions

This report deals with three broad fields of research related to each other: Artificial Intelligence, Data Science, and Smart Robotics. There is no single definition in the literature for these terms. The lack of a common definition was also demonstrated in the interviews conducted by the research team, in which various respondents expressed different opinions with regards to the definitions and the boundaries between them. In light of this finding, we chose to look at both artificial intelligence and data sciences at their broadest form. For example, artificial intelligence is defined as:

"Method for programming computers to enable them to carry out tasks or behaviors that would require intelligence if performed by humans"

National Academy of Sciences, 2018

The definition of Data Science is also complex. The relevant technologies comprising this scientific field are:

- **Technologies for collecting information**: Technologies that collect information for various uses, such as the Internet of Things (IoT).
- **Technologies for Information processing and analysis**: data mining technologies, information visualization and predictive analytics.
- **Blockchain-based technologies**: Blockchain Technologies, autonomous algorithms and smart contracts.

However, domains such as Analytical Prediction can also fall under the broad definition of Artificial Intelligence. Indeed, some interviewees referred to Data Science as the base of Artificial Intelligence, while others referred to Data Science as a field that includes Artificial Intelligence.

The term 'Smart Robotics' indicates a link between robots or actuators in the physical world and artificial intelligence. The EU attributes the following to 'Smart Autonomous Robots' (Nevejans, 2017):

- Acquires autonomy through sensors and/or by exchanging data with its environment (inter-connectivity) and trades and analyses data;
- is self-learning (optional criterion);
- has a physical support;
- adapts its behaviors and actions to its environment

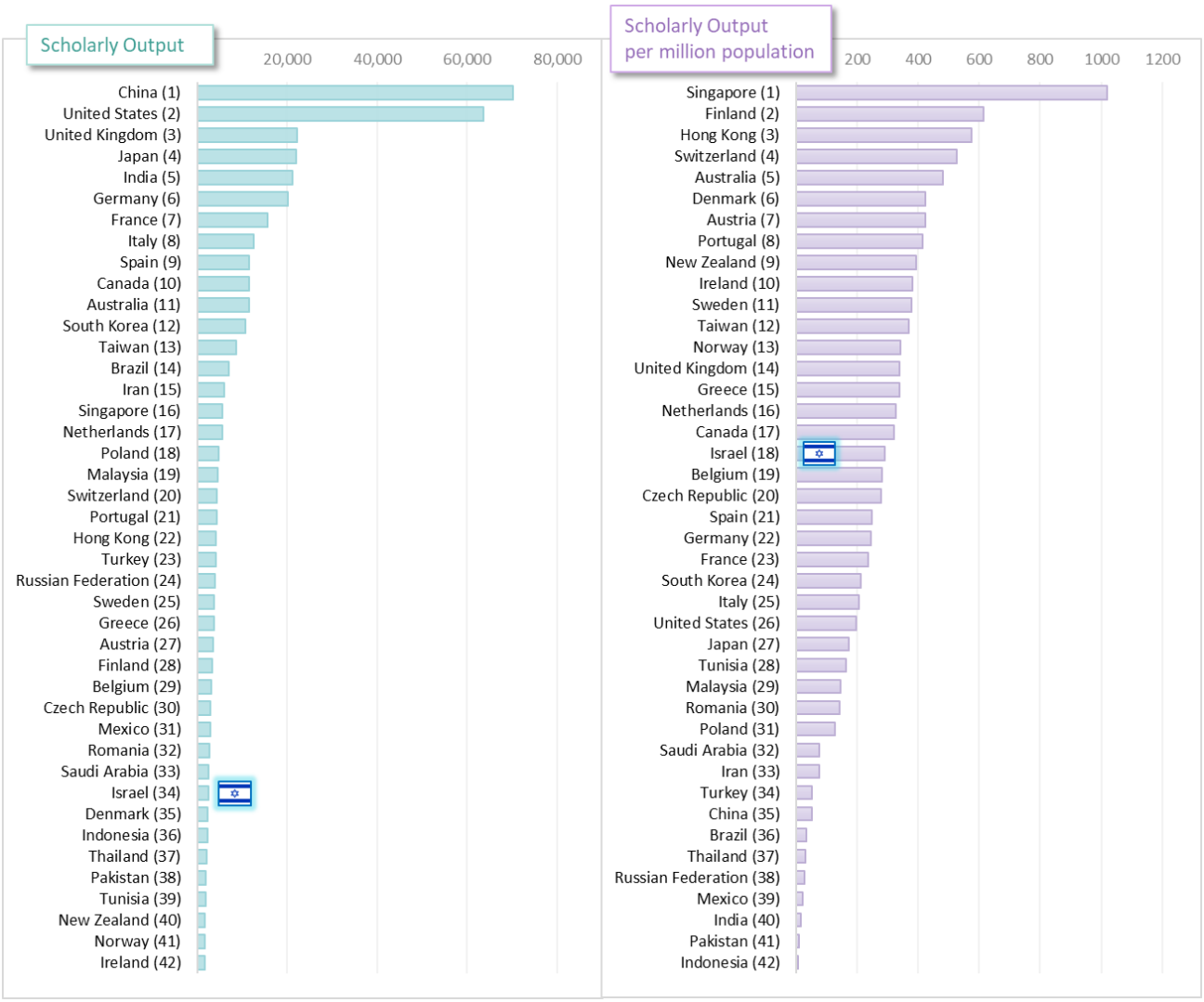
Research Outputs in Israel

► Scientific Publications

Preliminary findings show that China and the United States account for more than 36% of all world scientific publications in 2013-2017 in the categories defined as relevant to the research[1]. Israel is ranked 34th in total world scientific publications in these categories and 18th in publications per capita. (The leading countries in publications per capita are Singapore, Finland, Switzerland and Australia).

[1] The sub categories Artificial Intelligence, Human-Computer Interaction and Computer Vision and Pattern Recognition within the Computer Science category in the Elsevier's Scopus database.

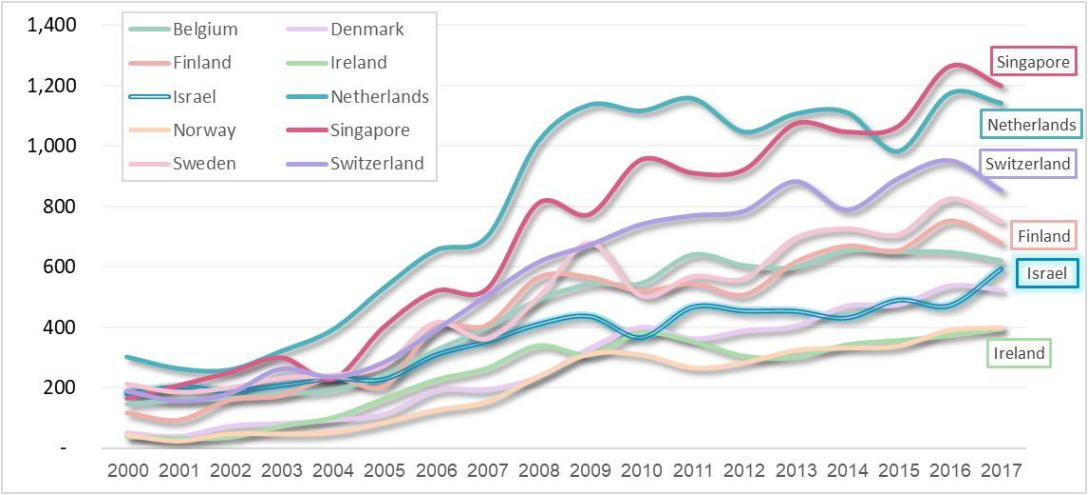
Figure 1: Total publications and publications-per-capita[2] by country, 2013-2017



Despite the increase in total Israeli publications between the years 2000-2017 (an increase of 229%), the rate of growth in countries such as Singapore, the Netherlands and Switzerland was higher (630%, 279% and 351%, respectively). The rate of growth in publications of the OECD countries was also larger than that of Israel in this period (409%).

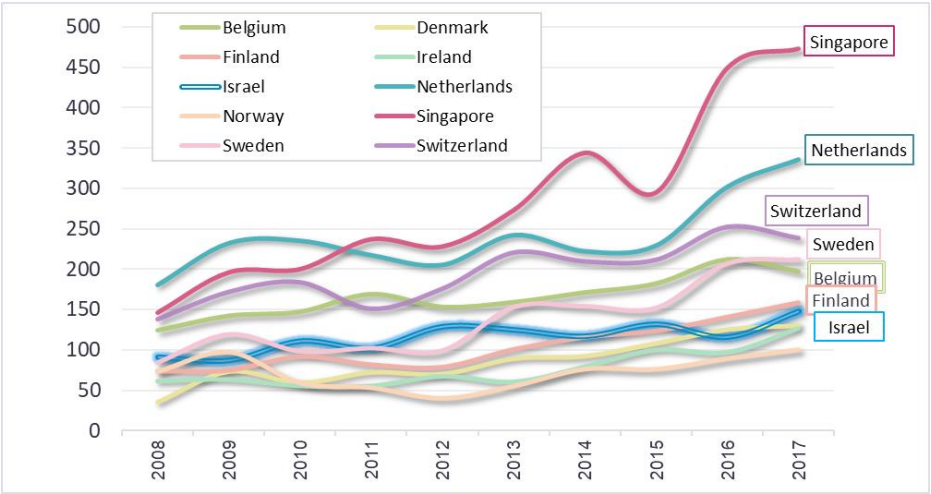
[2] In Artificial Intelligence (AI), Human-Computer Interaction (HC), Computer Vision and Pattern Recognition (CV)

Figure 2: Changes in the number of Israeli publications[2] compared with 'comparable countries'



High rate of growth in international cooperation between researchers contributes to the sharing of knowledge and the advancement of science. The number of Israeli joint publications has increased by 63% in years 2008-2017, but the growth rate was relatively low with respect to comparable-countries such as Singapore, which demonstrated a 224% increase in joint publications in this time period.

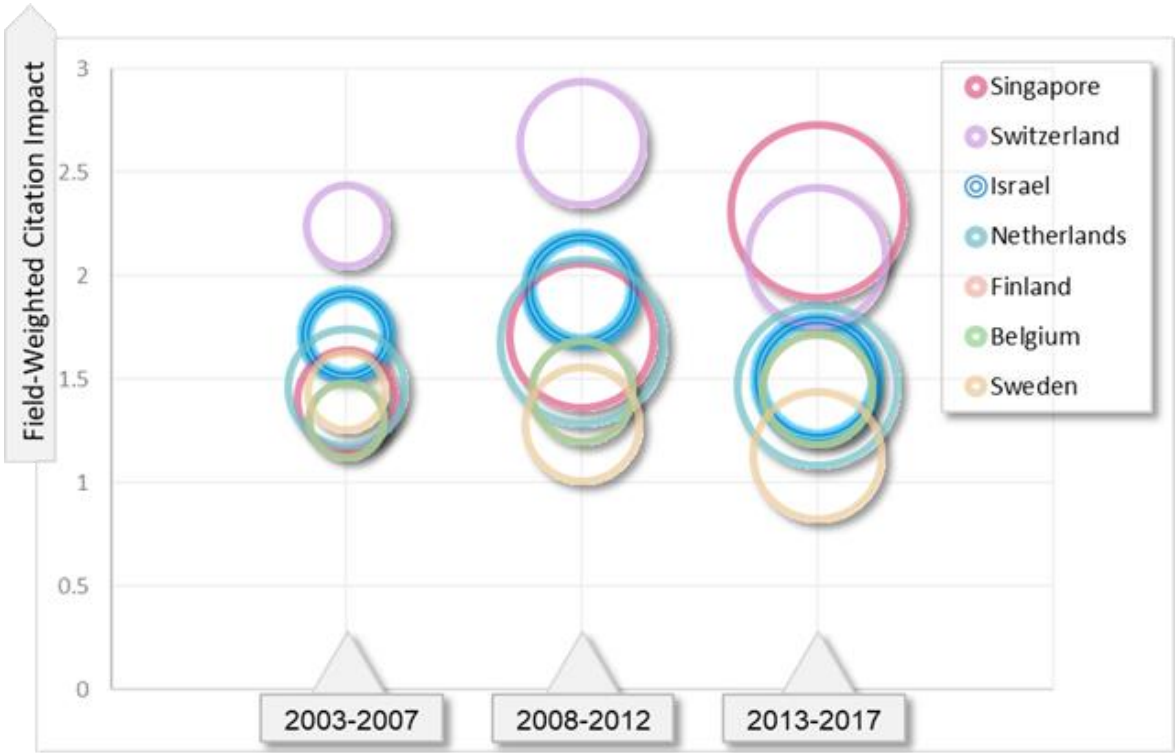
Figure 3: Number of internationally coauthored publications[3], 2008-2017



The number of citations per article constitutes a major indicator of the academic impact of the study. Singapore leads the growth rate in the number of publications in the group of the 10% most cited articles in the Artificial Intelligence category (252%). Israel demonstrated a moderate increase in this index in the relevant period.

[3] Artificial Intelligence category in Scopus/SciVal

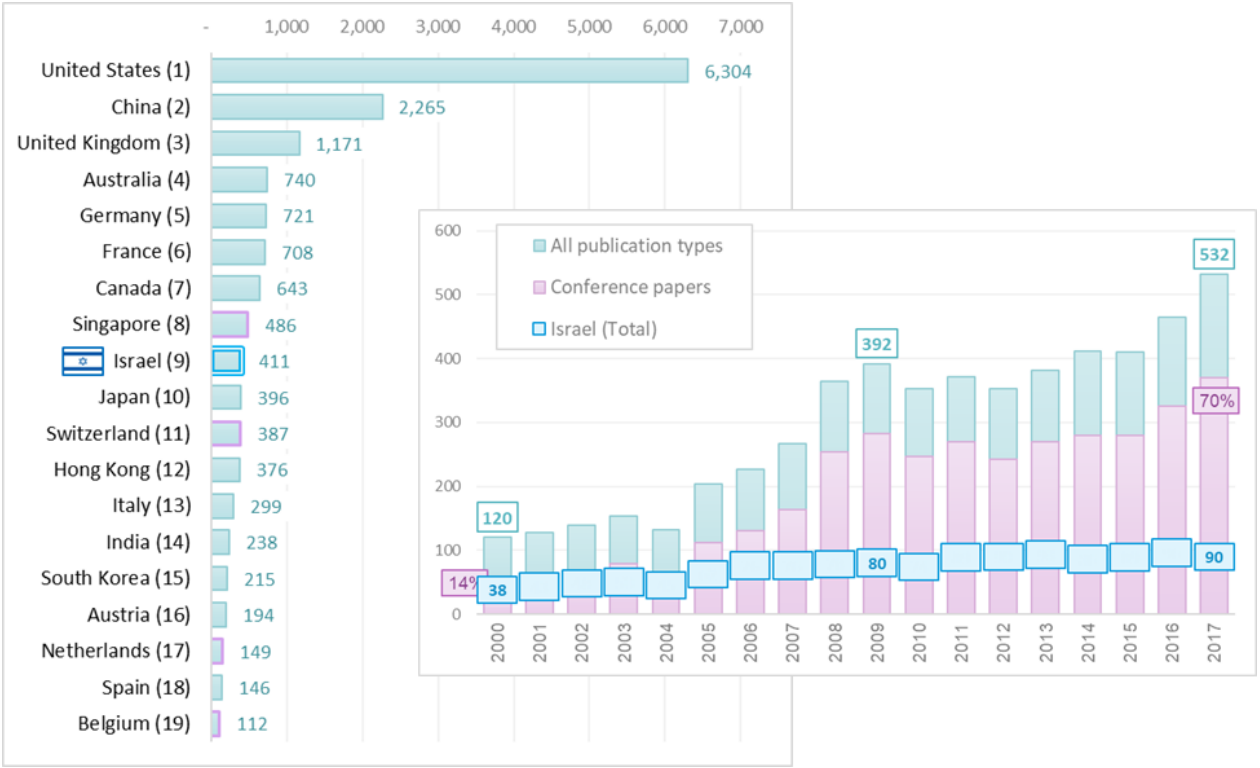
Figure 4: Number of publications[3] and normalized average citations in Israel and in 'comparable countries' [3], 2003-2017



It should be noted that the number of leading researchers who have published more than five articles in the last five years in the field of Artificial Intelligence in countries such as Singapore, the Netherlands and Belgium is significantly higher than the number of prominent researchers in this field in Israel. However, Israeli researchers do not fall short of researchers from these countries in terms of productivity (number of publications per researcher), and even surpass them in some cases.

Israel researchers published more scholarly output in 5 leading conferences (AAAI Conference on Artificial Intelligence, Computer Vision and Pattern Recognition, International Conference on Machine Learning, International Joint Conferences on Artificial Intelligence, Neural Information Processing Systems) in 2013-2017 than any other 'comparable country' except Singapore. Israel ranks 9 in the world in the number of publications in these conferences in the last 5 years.

Figure 5: Publications in 5 leading conferences by country, 2013-2017

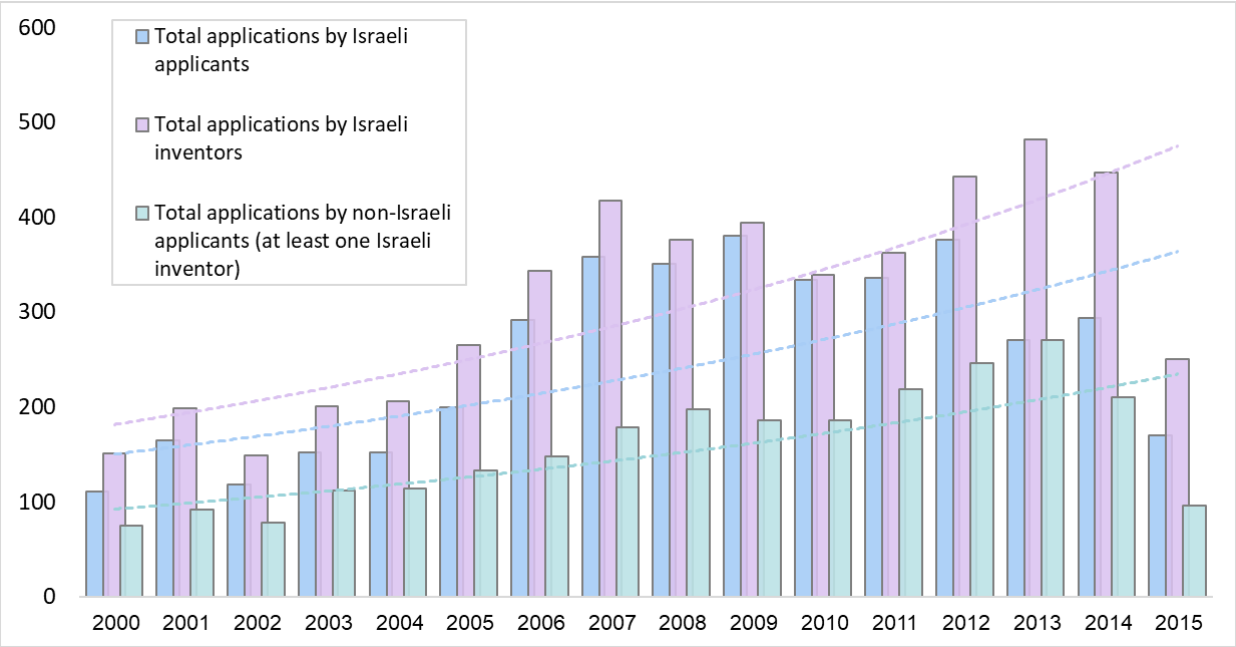


Israeli researchers have received 483 ERC grants, with only five of the EU member states and one associated state surpassing them in the number of grants. More than 5% (at least 26 grants) of the ERC grants received by researchers from Israeli universities were in fields related to Artificial Intelligence, Data Science and Smart Robotics.

► Patents

The total number of distinct applications for AI patents applied by Israeli applicants increased by an average of about 10% per year between the years 2000-2014. Applied Materials (Israel) leads the Israeli companies in the number of distinct patent applications, with IBM leading in number of patent applications by multinational companies with R&D centers in Israel.

Figure 6: Patent applications in AI for Israeli applicants, Israeli inventors and non-Israeli applicants (at least one Israeli inventor)



The leading patent technology classification (in terms of the number of distinct applications) is the 'Input arrangements for transferring data', which refers to input processing that can be manipulated by the computer (i.e. connected to a computer interface). This is also the leading technology for multinational companies with R&D centers in Israel, followed by a writing recognition technology.

Findings and Insights Based on In-Depth Interviews

The research team had conducted about sixty in-depth interviews, with several goals in mind:

- Understanding the current state of Artificial Intelligence, Data Science and Smart Robotics in Israel
- Qualitative analysis of the following research questions:
  - What are the main problems that practitioners in the relevant fields face? (e.g. manpower, training, research and infrastructure)
  - What are the fields of knowledge and business sectors in which Israel has an advantage within the broad fields of Artificial Intelligence, Data Science and Smart Robotics?
  - How can these fields be advanced in Israel?

Despite the relatively high number of interviewees, the findings in the next section should be regarded merely as tentative indicators. In the next part of the study, these indicators will be examined by surveying over 160 Israeli firms in the relevant fields.